Curriculum Vitae

Personal information

Name Solal Perrin-Roussel

Birthdate *4 August 1998*

Address ENS de Lyon, site Monod. UMPA UMR 5669 CNRS.
46, allée d'Italie, 69364 Lyon Cedex 07, France.

E-mail address solal.perrin-roussel@ens-lyon.fr

Education

2021 – 2022 **École Normale Supérieure Paris-Saclay** *Gif-sur-Yvette*
Master 2 Recherche at Université Sorbonne Paris Nord
M2 in fundamental mathematics.
Courses taken: Spectral Theory, Microlocal and semiclassical analysis and spectral asymptotics, Uncertainty Principles, Topological Data Analysis, Random Operators.

2020 – 2021 **École Normale Supérieure Paris-Saclay** *Gif-sur-Yvette*
Master 2 Formation à l'Enseignement Supérieur
Agrégation in mathematics.
Ranked 27th.

2019 – 2020 **Ecole Normale Supérieure Paris-Saclay (ENS)** *Cachan, France*
Master 1: Master's degree in mathematics
Courses taken : Algebra, Probabilities, Analysis, Geometry, Evolution Problems, Compact Groups and Lie Groups, Spectral Theory for Quantum Mechanics

2018 – 2019 **Ecole Normale Supérieure Paris-Saclay (ENS)** *Cachan, France*
Licence 3: Bachelor in mathematics
Courses taken : Algebra, Probabilities, Fourier and Hilbert Analysis, Integration, Ordinary Differential Equations, Differential Calculus, Complex Analysis, Partial Differential Equations, Quantum Mechanics

2015 – 2018 **Lycée Blaise Pascal** *Orsay, France*
Classe préparatoire aux grandes écoles: a selective undergraduate programme in the sciences preparing for the competitive entrance exam to the ENS and to engineering schools

2012 – 2015 **Lycée Gustave Monod** *Enghien-les-Bains, France*
Baccalaureate, Scientific section – Option Mathematics : Qualification "Very good"
Abibac (Double Bachelor's degree with Germany, including a 3 months-long exchange in 2012 in Wiesbaden)

Work experience

September 2025 - now **UMPA**, *ENS de Lyon, Lyon* – AGPR

- Agrégé préparateur (AGPR) at ENS de Lyon: Postdoctoral position with teaching duty.

September 2022 - September 2026 **CERMICS** *ENPC, Champs-sur-Marne* – PhD

- PhD title: Mathematical study of mean-field models for moiré materials.
  Directors: Éric Cancès and David Gontier

April – July 2022 **CEREMADE** *Université Paris-Dauphine PSL, Paris* – Research internship

- Topic : Semiclassical analysis in moiré materials – Supervisors : David Gontier (CEREMADE) and Éric Cancès (CERMICS, École des Ponts ParisTech)

May – August 2020 **CEREMADE** *Université Paris-Dauphine, Paris* – Research internship

- Topic: Topological invariant of the Kitaev chain – Supervisors : David Gontier (CEREMADE) and Domenico Monaco (Sapienza - Università di Roma)

April – June 2019 **CMLA** *ENS Paris-Saclay, Cachan, France* – Research internship

- Topic: Image denoising for a Cauchy noise – Supervisor : Agnès Desolneux

Teaching

2025-2026 **ENS de Lyon**, *Lyon*

- 1st semester: Tutorials of the Topology and Differential Calculus lecture by Laurent Berger (L3 students).
- 1st and 2nd semester: Preparation course for the Agrégation in mathematics (a master's-level competitive exam to become a mathematics teacher in France). Analysis lecture; Modeling and computational science lecture.

2024-2025 **ENS Paris** *rue d'Ulm, Paris*

- 1st semester: Supervision of a working group on Mathieu Lewin's book *Théorie spectrale et mécanique quantique* (M1 students).
- 2nd semester: Tutorials of the Optimization and Optimal Transportation lecture by Quentin Mérigot (L3 students).

2022-2024 **ENS Paris** *rue d'Ulm, Paris*

- 1st semester: Supervision of a working group on Mathieu Lewin's book *Théorie spectrale et mécanique quantique* (M1 students).
- 2nd semester: Tutorials of the Mathematical Modelling lecture by Bertrand Maury (L3 students).

Publications

June 2026 *preprint* Title: Kohn-Sham models for encapsulated two-dimensional materials. Co-author with Éric Cancès and David Gontier.

- ArXiv link: https://doi.org/10.48550/arXiv.2606.11785

June 2026 Title: A mathematical analysis of IPT-DMFT. Co-author with Éric Cancès et Alfred Kirsch.

- Communications in Mathematical Physics: https://doi.org/10.1007/s00220-026-05647-9

June 2025 *preprint* A mathematical analysis of the discretized IPT-DMFT equations. Co-author with Éric Cancès and Alfred Kirsch.

- ArXiv link: https://doi.org/10.48550/arXiv.2505.21287

April 2022 Title: Symmetric Fermi projections and Kitaev's table: topological phases of matter in low dimensions. Co-author with David Gontier and Domenico Monaco.
Published in **Journal of Mathematical Physics** (Vol.63, Issue 4) on April 21st 2022.

- Journal of Mathematical Physics: https://doi.org/10.1063/5.0084326

Conferences

April 2025 GAMM 2025, Poznań: presentation.

July 2024 ICMP 2024, Strasbourg: poster.
International workshop on 2D and moiré materials, Roscoff: presentation.

November 2024 Analysis team seminar at DMA, ENS Paris: presentation.

Languages

French, *Mother tongue*

English, *Cambridge Advanced, level C1, Date taken: 11/5/2019*

German, *Abibac, level C1, Date taken: 7/7/2015*

Italian, *Beginner*